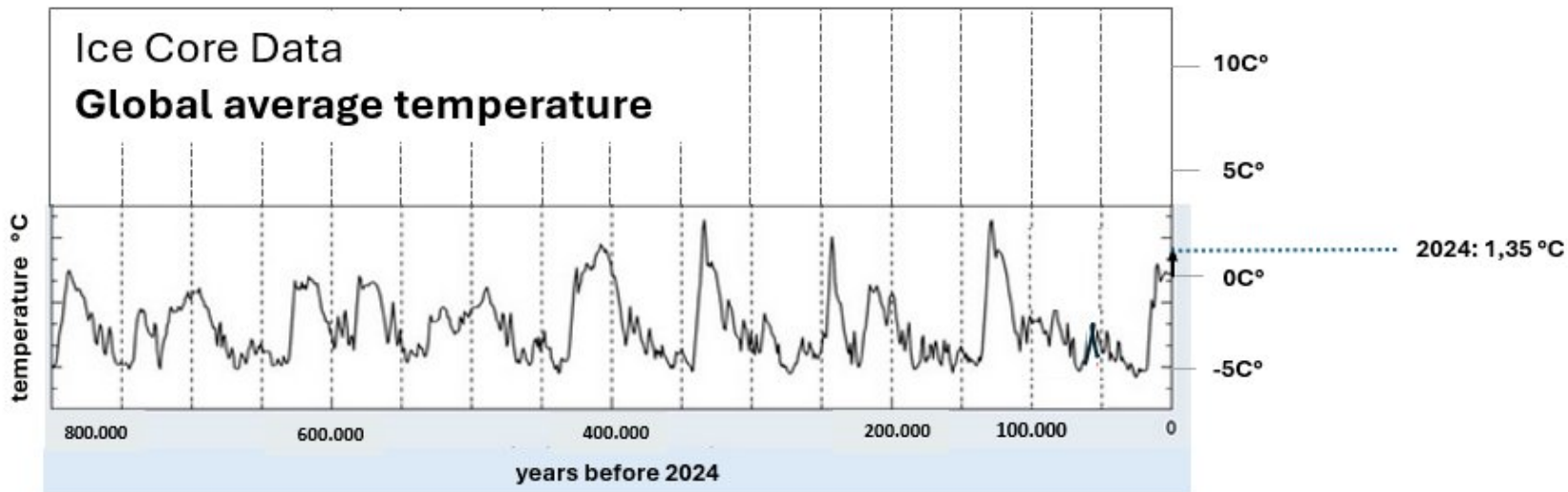
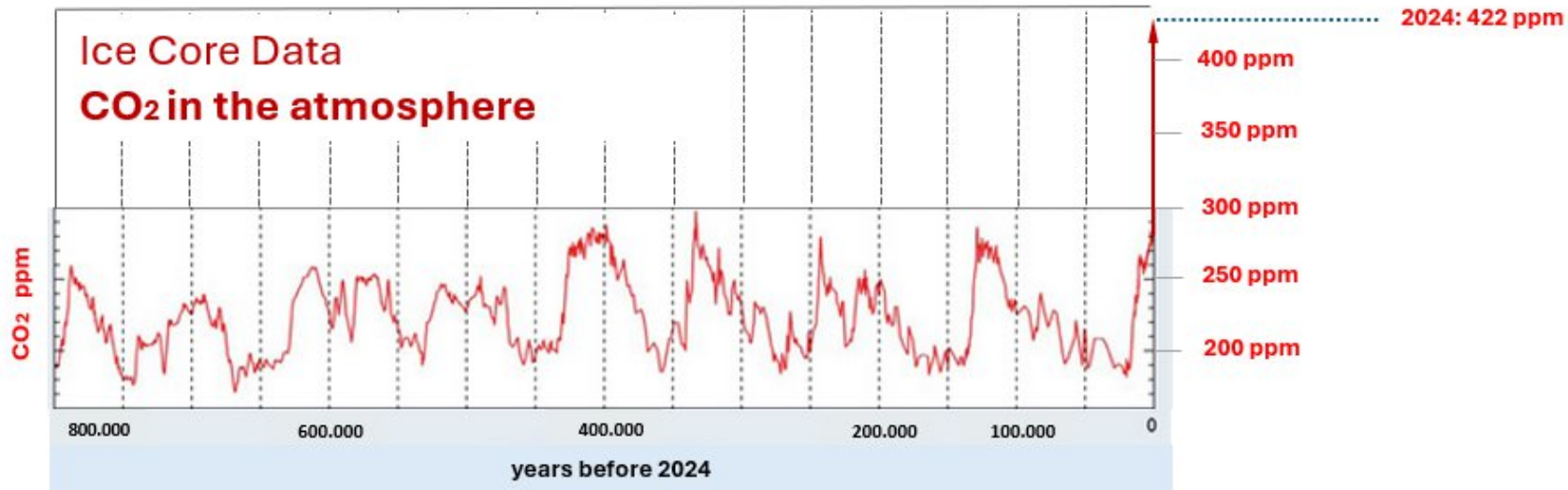




- The chart shows the level of CO₂ and the average temperature on our planet lasts 800.000 years .
- The global average temperature and the amount of CO₂ can be determined from the air bubbles in the ice of the antartica.
- 25 000 years before now, big parts of our continents were covered in ice. The global average temperature was about 6°C lower than today.
- To reach 300ppm, about 960 billion tons of CO₂ must be removed out of the air.
- 1ppm ~ 7,8 billion tons CO₂.

Chart: HTL Vöcklabruck
 Data: CO₂ and deuterium (temperature proxy) European Project for Ice Coring in Antarctica (EPICA)
www.antarctica.ac.uk
 Ice cores and climate change
https://bas.ac.uk/wp-content/uploads/2015/04/ice_cores_and_climate_change_briefing-sep10.pdf

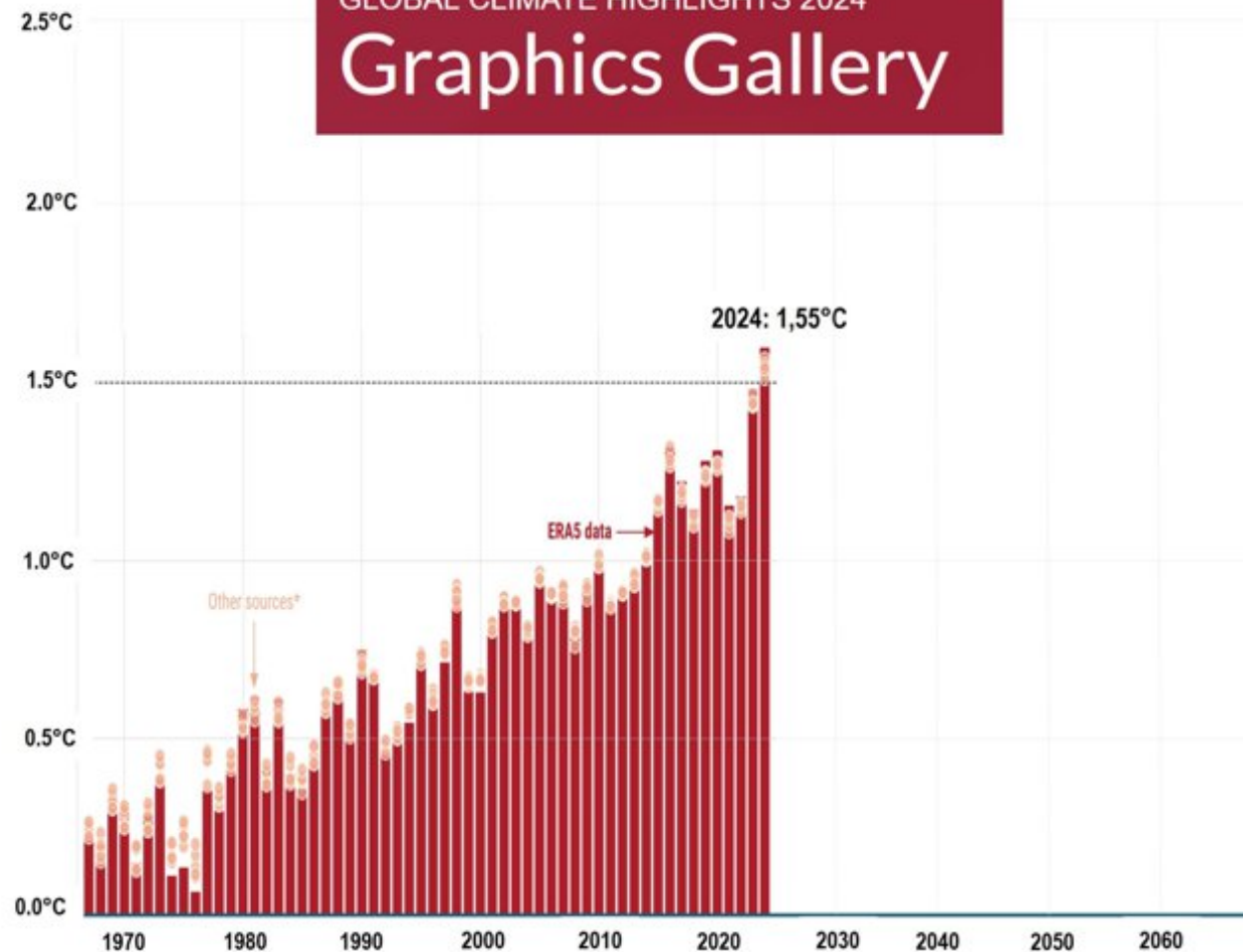
12.05.2025



Global surface temperature: increase above pre-industrial

GLOBAL CLIMATE HIGHLIGHTS 2024

Graphics Gallery



*Other sources comprise JRA-3Q, GISTEMPv4, NOAA GlobalTempv6, Berkeley Earth, HadCRUT5.

- In the year 2024 the increase of the global temperature was $\sim 1.55^{\circ}\text{C}$.

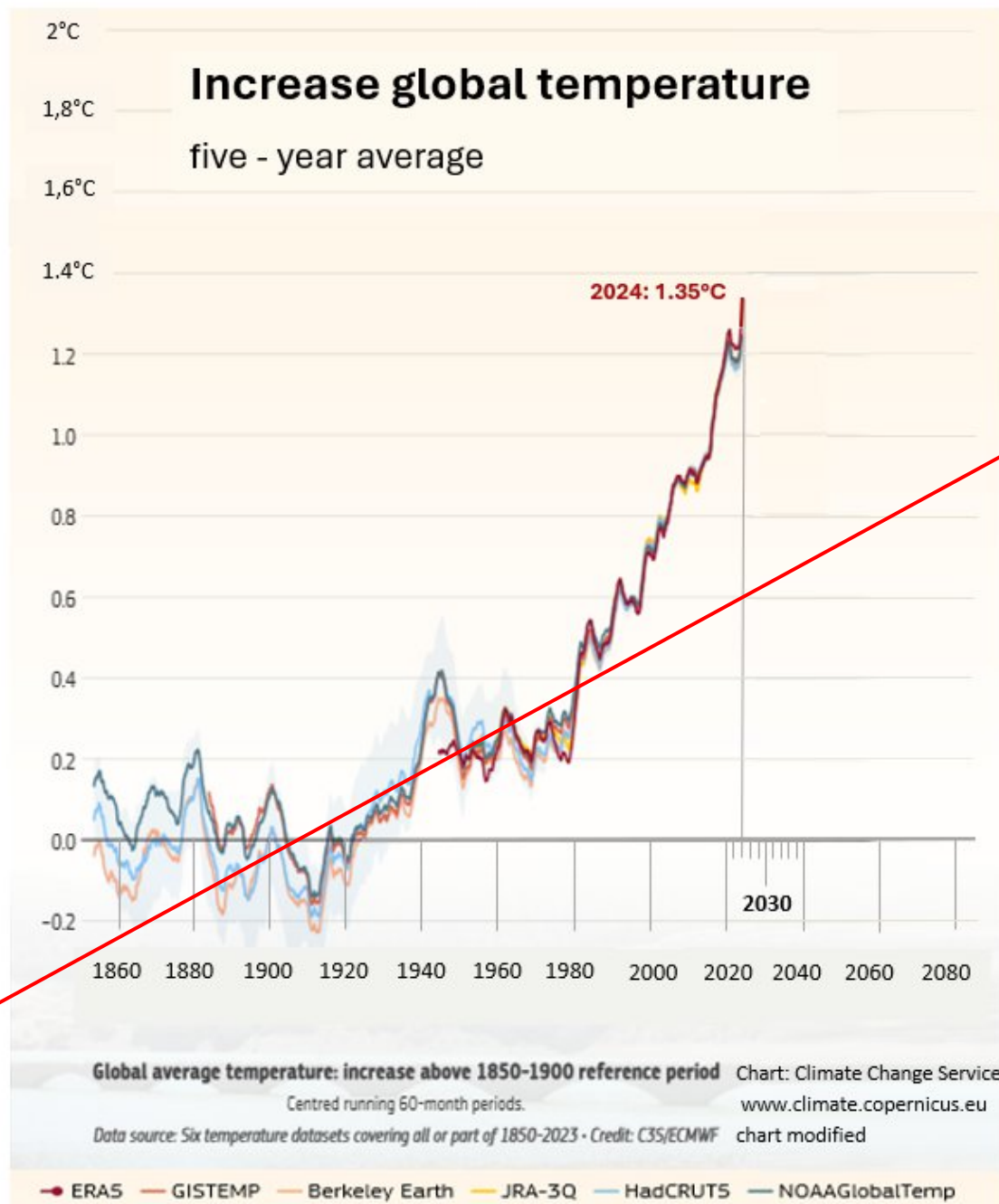
Source: Climate Change Service

climate.copernicus.eu 10.04.2025



PROGRAMME OF
THE EUROPEAN UNION



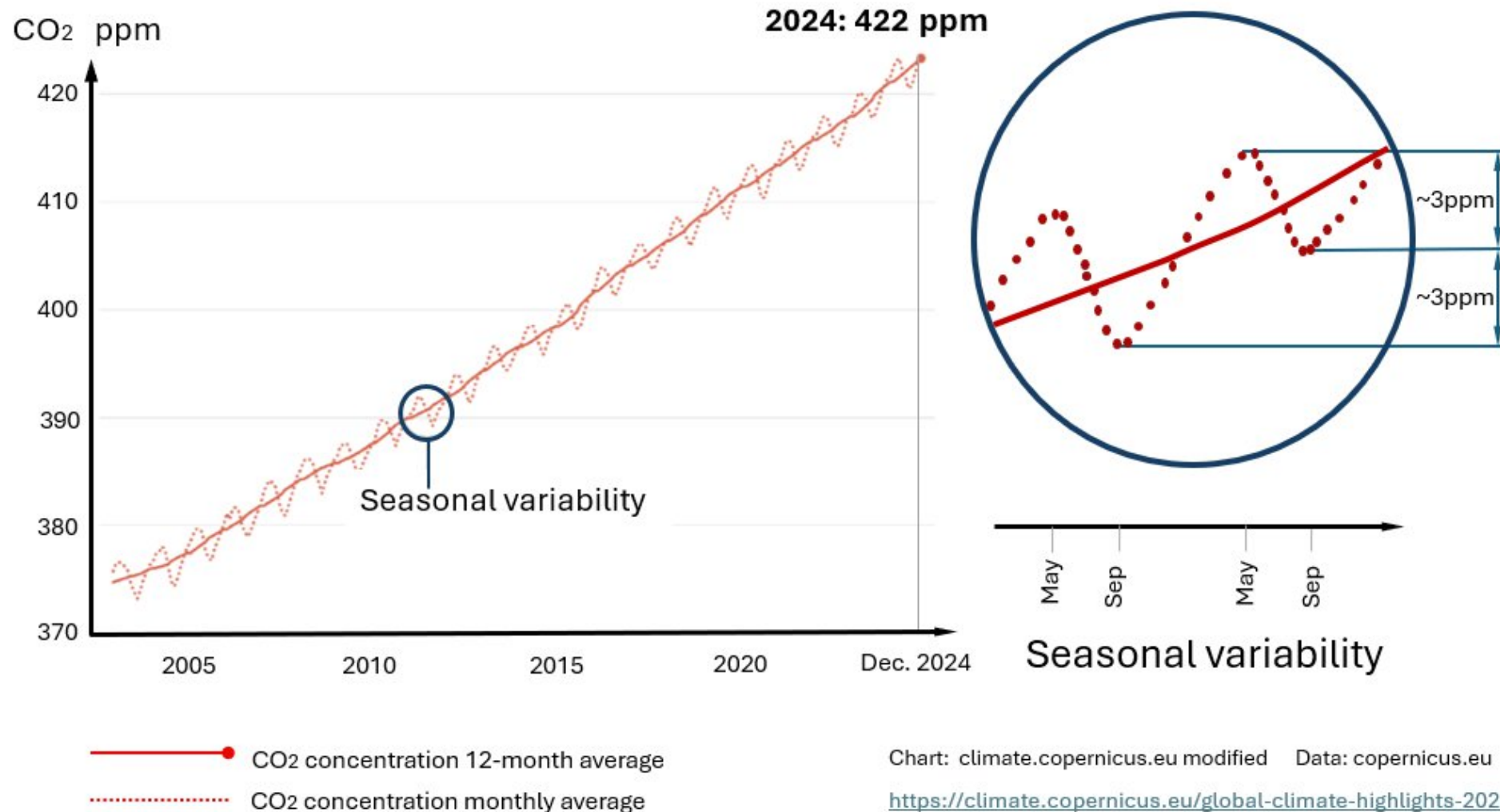


- In the year 2024 the increase of the global temperature was ~1.55°C.

Source: Climate Change Service

climate.copernicus.eu 10.04.2025

Increase of CO₂ in the atmosphere



- Each year humans emit about 40 billion tons of CO₂.
- 20 billion tons (~3ppm) are absorbed by plants (photosynthesis).
- The rest of CO₂ remains in the air.

Chart: climate.copernicus.eu modified Data: [copernicus.eu](https://climate.copernicus.eu)

<https://climate.copernicus.eu/global-climate-highlights-2024>

10.04.2025



Sea surface temperature anomalies

Data source: ERA5 • Reference period: 1991–2020 • Credit: C3S/ECMWF

Global (60°S–60°N)

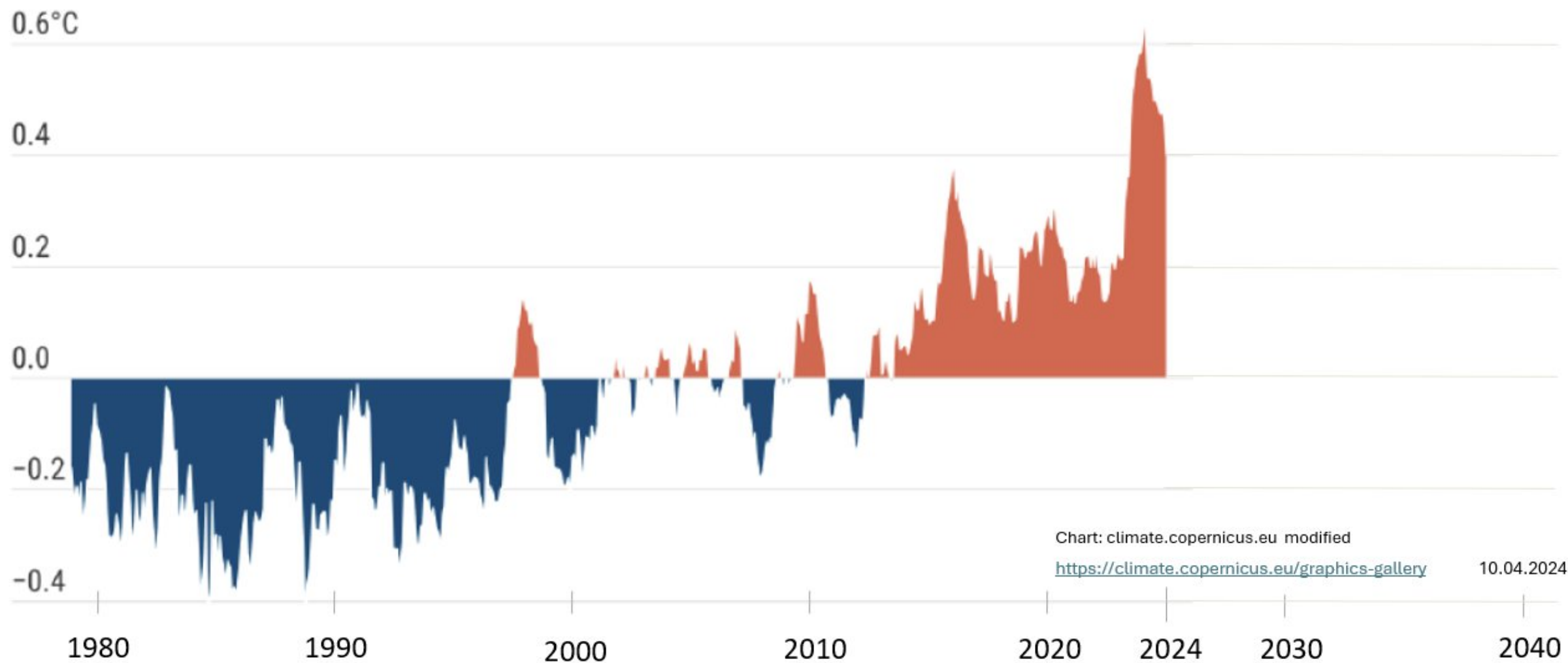


Chart: climate.copernicus.eu modified

<https://climate.copernicus.eu/graphics-gallery>

10.04.2024

CO₂ in the atmosphere reflects heat back to the earth.

More **CO₂** causes more reflection.

